

Richard Álvarez

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Professional Summary

Applied AI systems engineer building production agent systems, voice workflows, and enterprise integrations. Experienced in authorization, human approval controls, automated testing, observability, and reliable deployment across TypeScript and Python stacks.

Experience

- AI Solutions Engineer**
Layer 7 Systems | Independent Contractor

Sep 2025 – Present
Remote

 - Built a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph and Teams for structured service desk intake, ticket actions, and technician routing, then operated it in production for several months.
 - Engineered caller and company resolution, contact-safe ticket access, backend authorization gates, deterministic transfer eligibility, post-call workflows, and human fallback to prevent unsafe writes and cross-client data exposure.
 - Built an authenticated React operations dashboard end to end and expanded and maintained a 750+ test regression suite covering authorization, validation, transfers, external API behavior, alerts, and failure recovery.
 - Deployed Docker workloads through GitHub Actions across staging and production, then diagnosed duplicate tickets, stale identity state, API drift, retry and dead-letter queue failures, transfer defects, and CI and VPS incidents.
 - Built a workflow-analysis pipeline that reviewed 60 recordings from 14 users and identified 176 AI-assistable moments, then delivered a Next.js and Neon training platform with four courses and 16 modules for 34 trainees.
- Freelance Web Developer**
Richard Álvarez | Sole Proprietorship

Oct 2024 – Present
Chicago, IL

 - Design and ship responsive web applications for creative and commercial clients using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, and object storage.
 - Delivered two public-facing websites and continue developing several private web projects.
- IT Assistant**
Kenyon College Library & Information Services

Sep 2023 – Feb 2024
Gambier, OH

 - Supported workstation imaging, hardware deployment, printer and AV troubleshooting, and secure drive destruction, and built a spreadsheet tool to automate classroom maintenance scheduling.

Selected Project

- Vigil** | TypeScript, Bun, Hono, SQLite FTS5, Next.js, Cloudflare | GitHub

2026

 - Built an open-source, self-hosted email intelligence agent with multi-model routing, encrypted user-supplied credentials, persistent memory, deterministic action mapping, alerts, audit logs, and per-invocation cost tracking.

Education

- Kenyon College**
Bachelor of Arts in Film
Minor in History; Integrated Program in Humane Studies concentration

Aug 2020 – May 2024
Gambier, OH

Courses: Software Development, Data Structures and Programming Design, AI for the Humanities

Technical Skills

- Agent Systems:** MCP, tool calling, RAG, structured outputs, context and memory design, agent orchestration
- Workflow and Integration:** n8n, REST APIs, webhooks, Autotask, Microsoft Graph and Teams, identity and permissions, human approval controls
- Reliability and Delivery:** Docker, GitHub Actions, Linux and VPS, CI/CD, automated testing, validation, logging, observability, incident response, cost monitoring
- Product and Data:** TypeScript, Python, Bun and Node.js, Hono and FastAPI, React and Next.js, PostgreSQL, SQLite FTS5
- Systems and Media:** C++20, CMake, FFmpeg, PySceneDetect